Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3.

Multilingual support for template handling
Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation
Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types
Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers
TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

[T1]fontenc [utf8]inputenc [english]babel pgfplots compat=newest
booktabs siunitx
lmodern charter
[department=miljo]beamerthemeDTU

DTU Beamer Template

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19 Pebruari 2025

Outline

User Guide

- Package Options
- Date Format

Demonstration

- Lists
- Blocks
- Tables
- Plots
- Frame Numbers

Final Remarks

- Fonts

Package Options I

[department=]

• aqua	DTU Aqua
• byg	DTU Civil Engineering
• compute	DTU Compute
• elektro	DTU Electrical Engineering
• energikonvertering	DTU Energy Conversion
• fotonik	DTU Fotonik
• fysik	DTU Physics
• food	DTU Food
• kemi	DTU Chemistry
• kemiteknik	DTU Chemical Engineering

[department=]

Package Options II

- management — DTU Management Engineering
- mekanik — DTU Mechanical Engineering
- miljo — DTU Environment Engineering
- nanotek — DTU Nanotech
- space — DTU Space
- systembiologi — DTU Systems Biology
- transport — DTU Transport
- vaterinaerinstituttet — DTU Vet
- vindenergi — DTU Wind Energy

[showsection=]

- false — Remove section from above frame title
- true (default) — Display section above frame title

Date Format

Default Date Format

```
[Numeric]Day.[Numeric]Month.[Numeric]Year
```

Customize Date Format

Place the following code snippet in the preamble:

```
\newdateformat{myDateFormat}{\THEDAY-\THEMONTH-\THEYEAR}
\renewcommand{\DTUDateFormat}{\myDateFormat}
```

Lists

- Notice
- the
- red
- bullet

1. Wow
2. numbered
3. list

Blocks

Cool block

Get nice visual effects by organizing content into **blocks**. Title background color matches the red from DTU logo.

Tables

Tabel: Not a regular table. Content is aligned with respect to the decimal symbol.

Some Values
2.3456
34.2345
-6.7835
90.473
5642.5
1.2e3
e4

Plots

Stunt your colleagues with amazing plots (pgfplots).

[width=0.4grid=major, title=Model Validation, xlabel=X, ylabel=Y]
$-x^5 - 242$; *model*
coordinates (-4.77778,2027.60977) (-3.55556,347.84069)
(-2.33333,22.58953) (-1.11111,-493.50066) (0.11111,46.66082)
(1.33333,-205.56286) (2.55556,-341.40638) (3.77778,-1169.24780)
(5.00000,-3269.56775) ; estimate

Frame number instead of page number

- Watch the frame number.
- It doesn't change!

Frame number instead of page number

- Watch the frame number.
- It doesn't change!

Hello Blackness

Fonts

- You can set whatever font you like in the preamble. E.g.:
  - `\usepackage{lmodern} % Latin Modern`
  - `\usepackage{helvet} % Helvetica`
  - `\usepackage{verdana} % Verdana`
  - `\usepackage{newtx, newtxmath} % Times (text and math)`
- DTU recommends Verdana but this font type is not available by default on all operating systems:
  - Windows: include the `verdana` package in the preamble and MiKTeX will do the rest;
  - Linux and OS X: before including the `verdana` package in your preamble, run the `install_fonts.sh` script from our package.

[remember picture,overlay] [fill=black, fill opacity=0.9, text=white, text opacity=1.0, rounded corners=5pt, font=] at (current page.center)

Your Name
LaTeX Support Group, Technical University of Denmark (DTU)
Building 308, Room 119 latex-support@student.dtu.dk. ;
2800 Kgs. Lyngby, Denmark +45 4525 phone
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[remember picture,overlay] [fill=black, fill opacity=0.9, text=white, text opacity=1.0, rounded corners=5pt, font=] at (current page.center)

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Contact

File bugs and feature requests at
`http://gitlab.gbar.dtu.dk/latex/dtutemplates/issues`

Release History I

v1.12

- Frame number instead of page number.

v1.11

- Long titles are displayed correctly in the footer;
- Department logos have the correct RGB color;
- The aspect ratio of the slides can be changed using beamer options.

v1.1

Release History II

- Fonts options;
- Poster template;
- Frame backgrounds;
- Black frames with white logo;
- Address boxes.

v1.03

- New logos for `vindenergi` and `kemiteknik`.

v1.02

- Date format is now customizable.

v1.01

- `dtuletter` template included;
- Package name changed to `dtutemplates`.